

Course Project: Fine-Tuning LLMs for Text Generation

Project Overview

Each project can be done by either a single student or a group of two students. Each group or student will select **one dataset** and **one model** from the provided options. The objective is to fine-tune the chosen model for various text generation tasks and evaluate its outputs using a comprehensive set of metrics.

Datasets

- **IntentCONANv2**
Input: hatespeech, csType
Output: counterspeech
<https://huggingface.co/datasets/Aswini123/IntentCONANv2>
- **MultiCONAN**
Input: hatespeech, csType, Emotion
Output: counterspeech
<https://huggingface.co/datasets/Aswini123/MultiCONAN>
- **CONAN**
Input: hatespeech
Output: counterSpeech
<https://github.com/marcoguerini/CONAN/tree/master/CONAN>
- **DialoCONAN**
Input: Utterances except last one
Output: Last utterance
<https://github.com/marcoguerini/CONAN/tree/master/DIALOCONAN>
- **Dolly Dataset**
Input: instruction
Output: response
<https://huggingface.co/datasets/databricks/databricks-dolly-15k>
- **Dialogsum**
Input: dialogue
Output: summary
<https://huggingface.co/datasets/knkarthick/dialogsum>

- **SQuAD v1.1**
Input: context (passage), question
Output: answer (text span from context)
<https://www.kaggle.com/datasets/akashdesarda/squad-v11>
- **Alpaca Dataset**
Input: instruction, input (optional)
Output: output
<https://huggingface.co/datasets/tatsu-lab/alpaca>
- **CodeAlpaca**
Input: instruction
Output: output (code solution)
<https://huggingface.co/datasets/sahil2801/CodeAlpaca-20k>
- **Synthetic Text-to-SQL Dataset**
Input: Natural language question about a database
Output: SQL query
https://huggingface.co/datasets/gretelai/synthetic_text_to_sql

Model Choices

- Gemma-2B
- Flan T5 Large
- Llama-2-1.3B
- Mistral-1.3B
- GPT2 Large
- GPT-Neo 1.3B
- Qwen2-0.5B-Instruct
- LaMini-GPT 1.5B
- Phi-2
- Pythia 1.4B

Project Steps

1. **Dataset and Model Selection**
 - Choose **one** dataset from the dataset list above.
 - Choose **one** model from the model list above.
2. **Data Preparation**

- Preprocess the dataset to extract required input and output columns.
- For IntentCONANv2: Use `hatespeech` and `csType` as input, `counterspeech` as output.

3. Model Fine-Tuning

- Set up the selected model in a Google Colab or a Kaggle environment.
- **Along with the full fine-tuning, apply parameter-efficient fine-tuning methods (LoRA, QLoRA, and Prefix-Tuning) in Colab's or Kaggle resource constraints.**
- Train the model using the prepared dataset.

4. Text Generation

- Use the finetuned model with different setups to generate counter-speech responses for a held-out test set.
- Ensure consistent input formatting for evaluation.

5. Evaluation Metrics

- Assess the generated text using the following metrics:
 - BLEU
 - ROUGE-1, ROUGE-2, ROUGE-L
 - METEOR
 - GLEU
 - Repetition Rate
 - Flesch Reading Ease
 - CoSIM
 - BERT Score
 - Toxicity
 - Novelty
 - Diversity

6. Results & Analysis

- Present the metric scores in a clear table with different Parameter-efficient setups vs the Full fine-tuning setup.
- Analyze strengths and weaknesses of the model's generated outputs.
- Discuss trade-offs (e.g., fluency vs. diversity, toxicity reduction, etc.).

7. Report Submission

- Your final report should include:
 - **Introduction:** Brief overview of the task and chosen setup.
 - **Methodology:** Dataset and model details, preprocessing, training approach.
 - **Evaluation:** Metric results, tables, and qualitative analysis.
 - **Conclusion:** Key findings and possible improvements.

Example Evaluation Table

Metric	Score
BLEU	
ROUGE-1	
ROUGE-2	
ROUGE-L	
METEOR	
GLEU	
Repetition Rate	
Flesch Reading Ease	
CoSIM	
BERT Score	
Toxicity	
Novelty	
Diversity	

Table 1: Evaluation metrics for text generation.

Additional Guidelines

- The code should be written in one notebook file, all the dependencies like downloads and others should be done internally in such a way that during evaluation, the code should run smoothly without any fail. (Penalties will be imposed in the case of failure.)
- Ensure reproducibility by documenting all code and hyperparameters.
- Clearly state any limitations or challenges encountered during the project in the final project report.